

Width N (bits)	Minimum value	Maximum value if nullable	Maximum value if not nullable
56	0 (0x0000000000000000)	72057594037927934 (0xFFFFFFFFFFFFFFFE)	72057594037927935 (0xFFFFFFFFFFFFFFF)
64	0 (0x0000000000000000)	18446744073709551614 (0xFFFFFFFFFFFFFFFE)	18446744073709551615 (0xFFFFFFFFFFFFFFF)

7.19.1.4. Signed Integer Type (8, 16, 24, 32, 40, 48, 56 and 64-bit)

This type represents an signed integer with length of N bits and a usable range of:

- $[-(2^{(N-1)}) \dots 2^{(N-1)} - 1]$ if not nullable OR
- $[-(2^{(N-1)} - 1) \dots 2^{(N-1)} - 1]$ if nullable.

Whether to use two's complement or another representation for the implementation of the Data Model is a consideration of the underlying encoding specification.

The following table presents the representable values in base-10 following the above rules:

Width N (bits)	Minimum value if nullable	Minimum value if not nullable	Maximum value
8	-127	-128	127
16	-32767	-32768	32767
24	-8388607	-8388608	8388607
32	-2147483647	-2147483648	2147483647
40	-549755813887	-549755813888	549755813887
48	-140737488355327	-140737488355328	140737488355327
56	-36028797018963967	-36028797018963968	36028797018963967
64	-9223372036854775807	-9223372036854775808	9223372036854775807

7.19.1.5. Single-Precision Type

The single precision number format is based on the [IEEE 754-2019](#) single precision (32-bit) format for binary floating-point arithmetic.

See [IEEE 754-2019](#) for more details on the representable values.

7.19.1.6. Double Precision Type

The double precision number format is based on the [IEEE 754-2019](#) double precision (64-bit) format for binary floating-point arithmetic.

The format and interpretation of values of this data type follow the same rules as given for the single precision data type, but with wider mantissa and exponent ranges.